

Lesson 4-3: Multiply and Divide Rational Expressions

Math Objective	Use structure of rational expressions to rewrite them in different forms. Understand that rational expressions are analogous to rational numbers and follow similar rules when multiplying and dividing.
Language Objective	Students will explain how to state the domain of a simplified rational expression using sentence frames.
Essential Understanding	I can find the product and the quotient of rational expressions.

1. Quick Check

Look at the expression: $\frac{x-2}{x-2}$	Do you think it always equals 1? Circle one:		
	Yes	No	Not sure

2. Test It

x	$\frac{x-2}{x-2}$
0	
1	
2	
3	
4	

3. Explain What You See

The expression **breaks** at _____ because dividing by zero is not allowed.

It equals 1 for all values **except** _____ because the denominator becomes zero.

Some values make an expression “break.” Now we will practice rewriting expressions and checking which values are allowed.

EXAMPLE 1 – Write Equivalent Rational Expressions

We can rewrite an expression by multiplying by a form of 1.

A *form of 1* looks like this:

$$\frac{x}{x} \qquad \frac{x+2}{x+2} \qquad \frac{5}{5}$$

These fractions all equal 1 (as long as the denominator is not zero).

Start with the expression:

$$x + 3$$

Multiply by a form of 1:

$$(x + 3) \cdot \frac{x}{x}$$

Rewrite it as a rational expression:

$$\frac{x(x+3)}{x}$$

These two expressions look different, but they represent the same value for all x except 0.

- The original expression $x + 3$ is defined for every real number.
- The rewritten expression $\frac{x(x+3)}{x}$ is undefined when $x = 0$ because the denominator becomes zero.

So the expressions are equivalent only on the shared domain.

- **Turn and Talk:** Why can multiplying by 1 change the domain?

Word bank: value — zero — denominator — restriction

TRY IT 1

Rewrite the expression below by multiplying by a form of 1. Then state the value that must be excluded from the domain.

$$x - 5 \longrightarrow (x - 5) \cdot \frac{x + 2}{x + 2}$$

Domain restriction: _____

EXAMPLE 2 – Simplifying a Rational Expression

A rational expression is in **simplest form** when the numerator and denominator have **no common factors** (other than 1).

We can simplify by **canceling common factors**, but we must also state the **domain** (values that make the denominator zero).

Simplify the expression:

$$\frac{(x-2)(x+5)}{(x-2)(x-3)}$$

Step 1: Identify the common factor

Both numerator and denominator contain the factor $(x - 2)$.

Step 2: Cancel the common factor

$$\frac{\boxed{(x-2)}(x+5)}{\boxed{(x-2)}(x-3)} = \frac{x+5}{x-3}$$

Step 3: State the domain

The original denominator was $(x - 2)(x - 3)$. So the expression is undefined when:

- $x = 2$
- $x = 3$

Final Answer

$$\frac{x+5}{x-3}, \quad \text{for } x \neq 2, 3$$

✱ **Short Write:** Explain why we must use the **original** denominator (not the simplified one) to determine the domain of a rational expression.

We use the original denominator because _____ . Even though the factor was canceled, it still affects the domain since _____ .

The simplified expression does not show the restriction at _____ because _____ . This shows that simplifying does not change _____ .

Try It 2

Simplify the expression and state the domain:

$$\frac{(y+4)(y-1)}{(y-1)(y-6)}$$

Simplified form: _____

Domain: $y \neq \underline{\hspace{1cm}}$, $y \neq \underline{\hspace{1cm}}$

EXAMPLE 3 – Multiplying Rational Expressions

When we multiply rational expressions, we multiply **straight across** (numerator $\times$ numerator, denominator $\times$ denominator). Then we **cancel any common factors**.

This works exactly like multiplying numerical fractions.

Multiply and simplify:

$$\frac{x}{x+3} \cdot \frac{x+3}{x-1}$$

Step 1: Multiply across

$$\frac{x(x+3)}{(x+3)(x-1)}$$

Step 2: Cancel the common factor

$$\frac{\boxed{(x+3)} \cdot x}{\boxed{(x+3)}(x-1)} = \frac{x}{x-1}$$

Step 3: State the domain

The original denominators were $x+3$ and $x-1$. So the expression is undefined when:

- $x = -3$
- $x = 1$

Final Answer

$$\frac{x}{x-1}, \quad \text{for } x \neq -3, 1$$

- **Turn and Talk:** Did canceling remove the domain restriction? Why or why not?

Word bank: *cancel — denominator — restriction — factor — zero — original expressions*

Try It 3

Multiply and simplify. Then state the domain.

$$\frac{x-4}{x+2} \cdot \frac{(x+2)(x+5)}{(x-4)(x-1)}$$

Simplified form: _____

Domain: $x \neq$ ____, $x \neq$ ____, $x \neq$ ____

EXAMPLE 5 – Dividing Rational Expressions

To divide rational expressions, we **multiply by the reciprocal** of the divisor. Then we cancel any common factors and state the domain.

Divide and simplify:

$$\frac{(x+1)(x+4)}{x-2} \div \frac{x+1}{(x-2)(x+5)}$$

Step 1: Multiply by the reciprocal

$$\frac{(x+1)(x+4)}{x-2} \cdot \frac{(x-2)(x+5)}{x+1}$$

Step 2: Cancel common factors

Cancel $(x+1)$ and $(x-2)$:

$$\frac{\boxed{(x+1)}(x+4)}{\boxed{(x-2)}} \cdot \frac{\boxed{(x-2)}(x+5)}{\boxed{(x+1)}}$$

What remains:

$$(x+4)(x+5)$$

Step 3: State the domain

The original denominators were:

- $x-2$
- $(x-2)(x+5)$

So the expression is undefined when:

- $x=2$
- $x=-5$

- $x = -1$ (from the canceled $(x + 1)$ in the divisor)

Final Answer

$$(x + 4)(x + 5), \quad \text{for } x \neq -5, -1, 2$$

✱ **Short Write:** Explain why dividing rational expressions can create multiple domain restrictions, including some that come from factors that get canceled.

Division creates more restrictions because _____. *Even canceled factors matter because they were originally in a _____.*

Any factor in the original denominators causes a restriction when _____. *This means the final domain must include restrictions from _____.*

Try It 5

Divide and simplify. Then state the domain.

$$\frac{(x - 3)(x + 2)}{x + 4} \div \frac{x - 3}{(x + 4)(x - 1)}$$

Simplified form: _____

Domain: $x \neq$ ____, $x \neq$ ____, $x \neq$ ____

CONCEPT SUMMARY Products and Quotients of Rational Expressions

	Multiply	Multiply an Integer or a Polynomial	Divide
RATIONAL EXPRESSIONS	$\frac{3x}{x+1} \cdot \frac{x^2+x}{3x-6}$ The domain is $x \neq -1$ or 2.	$\frac{x+2}{x^2-4} \cdot (x^2-2x)$ $= \frac{x+2}{x^2-4} \cdot \frac{x^2-2x}{1}$ The domain is $x \neq -2$ or 2.	$\frac{1-x^2}{x^2+3x-4} \div \frac{x+1}{x+4}$ $= \frac{1-x^2}{x^2+3x-4} \cdot \frac{x+4}{x+1}$ The domain is $x \neq -4, -1, \text{ or } 1$.
WORDS	Identify common factors and simplify.	Write the polynomial as a rational expression with 1 in the denominator. Then multiply.	Multiply by the reciprocal of the divisor.

Fraction Rules	
Adding Fractions with Common Denominators	$\frac{A}{B} + \frac{C}{B} = \frac{A+C}{B}$
Subtracting Fractions with Common Denominators	$\frac{A}{B} - \frac{C}{B} = \frac{A-C}{B}$
Adding Fractions with Different Denominators	$\frac{A}{B} + \frac{C}{D} = \frac{AD}{BD} + \frac{BC}{BD} = \frac{AD+BC}{BD}$
Subtracting Fractions with Different Denominators	$\frac{A}{B} - \frac{C}{D} = \frac{AD}{BD} - \frac{BC}{BD} = \frac{AD-BC}{BD}$
Multiplying Fractions	$\frac{A}{B} \times \frac{C}{D} = \frac{A \times C}{B \times D}$
Dividing Fractions	$\frac{A}{B} \div \frac{C}{D} = \frac{A}{B} \times \frac{D}{C} = \frac{A \times D}{B \times C}$

Select Exercises

- 12. Error Analysis** Describe the error a student made in multiplying and simplifying

$$\frac{x+2}{x-2} \cdot \frac{x^2-4}{x^2+x-2}$$

$$\begin{aligned} & \frac{x+2}{x-2} \cdot \frac{x^2-4}{x^2+x-2} \\ &= \frac{x+2}{\cancel{x-2}} \cdot \frac{(x+2)\cancel{(x-2)}}{(x+2)(x-1)} \\ &= \frac{2}{-1} \end{aligned}$$

DOK 2

Simplify the expression and state the domain.

$$\frac{(x-4)(x+7)}{(x-4)(x-1)}$$

DOK 1

Multiply, simplify, and state the domain.

$$\frac{x+3}{(x-2)(x+5)} \cdot \frac{(x-2)(x-1)}{x+3}$$

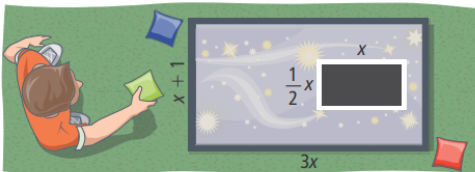
DOK 1

Divide, simplify, and state the domain.

$$\frac{(x-6)(x+2)}{x+4} \div \frac{x-6}{(x+4)(x-3)}$$

DOK 1

- 36. Model With Mathematics** Lupe designed a carnival game that involves tossing a beanbag into the box shown. In order to win a prize, the beanbag must fall inside the black rectangle. The probability of winning is equal to the ratio of the area of the black rectangle to the total area of the face of the box shown. Find the ratio that represents this probability in simplified form.



DOK 3